

Quality Inspection Report: Selective Laser Melting Process for 316L Stainless Steel Flat Washers

Report Number: QIR-SLM-316L-2023-08-015

Date of Inspection: August 15, 2023

Inspector: Maria Chen, Senior Quality Engineer

Facility: Advanced Manufacturing Division, Building C

Process Evaluated: Selective Laser Melting (SLM) of 316L Stainless Steel Flat Washers

Batch Reference: SLM-WSH-316L-230815-33

1.0 Executive Summary

This quality inspection report documents the verification of manufacturing processes and output quality for Batch SLM-WSH-316L-230815-33, comprising 33 units of 316L stainless steel flat washers produced via Selective Laser Melting. The inspection covered material handling, process parameters, energy utilization, and final product quality against internal specification AM-QS-316L-Rev4 and customer requirements.

All process parameters operated within established control limits. The manufactured washers demonstrated full compliance with dimensional and material specifications. Material utilization efficiency measured at 14.8% for finished product yield, with 71.5% of input powder successfully recovered for reuse. Energy consumption patterns aligned with equipment specifications and historical baselines.

Overall Assessment: PROCESS APPROVED - All quality parameters within acceptable limits.

2.0 Inspection Scope and Objectives

2.1 Scope of Inspection

This inspection evaluated the complete manufacturing cycle for 316L stainless steel flat washers, including: - Raw material verification and handling procedures - Water atomization process for powder production - Selective Laser Melting operational parameters - Shielding gas management and consumption - Product dimensional and visual quality - Waste stream management and material recovery

2.2 Quality Standards Reference

- Internal Quality Standard: AM-QS-316L-Rev4
 - Material Specification: ASTM A276/A276M-17 (316L Stainless Steel)
 - Additive Manufacturing Process: ISO/ASTM 52900:2021
 - Dimensional Inspection: ASME Y14.5-2018
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3.0 Materials and Process Verification

3.1 Material Consumption and Tracking

The material flow throughout the manufacturing process was meticulously documented and verified against batch records. All material movements were tracked using the facility’s material traceability system (MTS-AM v3.2).

Material Category	Specification	Quantity	Unit	Verification Method
Base Material	X2CrNiMo1712 Stainless Steel (316L equivalent)	4.11	kg	Certified Material Certificate #CMC-316L-4821
Process Water	Deionized Water for Atomization Cooling	16.8	kg	Flow Meter Calibration Cert #FMC-2284
Shielding Gas	High-Purity Argon (99.998%)	3.08	kg	Gas Mass Flow Controller Reading

Note: Argon consumption calculated based on chamber volume and component processing requirements.

3.2 Process Water Management

The water atomization process utilized 16.8 kg of process water for cooling during powder production. Water recovery systems successfully captured and treated 16.4 kg for reuse in subsequent cycles, representing a 97.6% recovery rate.

Historical Comparison: Previous quarter average recovery rate: 96.2% ±1.3%

4.0 Energy Utilization Analysis

4.1 Electrical Consumption

Energy consumption was monitored throughout the manufacturing cycle using calibrated power analyzers (Fluke 437 Series II). All measurements were verified against equipment specifications.

Process Stage	Equipment	Energy Consumption	Unit	Power Rating
Powder Production	Water Atomization Unit	2.55	kWh	25 kW nominal
Additive Manufacturing	SLM System (EOS M 290)	64.92	kWh	Laser On: 5.5 kW, Standby: 3.5 kW

4.2 Process Efficiency Metrics

The SLM process operated for 13.38 hours to complete the 33-component build cycle. Energy consumption per component averaged 1.97 kWh, consistent with equipment performance specifications.

Industry Benchmark: Typical SLM energy consumption ranges from 1.8-2.2 kWh per similar component in industry surveys.

5.0 Product Quality Assessment

5.1 Final Product Specifications

The batch produced 33 flat washers meeting all specified requirements. Product verification included dimensional inspection, visual examination, and material composition confirmation.

Product Parameter	Specification	Measured Value	Compliance
Total Product Mass	0.58-0.62 kg (33 units)	0.61 kg	PASS
Individual Unit Count	33 units	33 units	PASS
Material Composition	316L Stainless Steel	Verified via XRF	PASS
Surface Finish	Ra ≤ 12 μm	Ra 8-11 μm	PASS

5.2 Dimensional Verification

All 33 washers underwent comprehensive dimensional inspection using coordinate measuring machine (CMM) Mitutoyo CRYSTA-Apex S 121210. Critical dimensions including outer diameter, inner diameter, and thickness were within specified tolerances of ±0.1 mm.

Sample Measurement Data (5 units selected randomly):

Washer ID	Outer Diameter (mm)	Inner Diameter (mm)	Thickness (mm)	Status
WSH-001	20.08	10.52	2.05	PASS
WSH-008	20.05	10.49	2.03	PASS
WSH-015	20.11	10.51	2.06	PASS
WSH-022	20.07	10.48	2.04	PASS
WSH-029	20.09	10.53	2.05	PASS

6.0 Material Recovery and Waste Management

6.1 Powder Recovery System

The SLM process incorporates an integrated powder recovery system that separates reusable powder from process by-products. Recovery efficiency was verified through mass balance calculations.

Recovery Stream	Quantity	Unit	Disposition
316L Powder Reused in SLM	2.94	kg	Returned to process hopper
316L Powder for Remelting	0.15	kg	Sent to water atomization unit
Process Water Recovery	16.4	kg	Treated and returned to reservoir

6.2 Waste Generation and Disposal

Non-recoverable materials were properly segregated and disposed of according to facility waste management procedures.

Waste Category	Quantity	Unit	Disposal Method
Solid Waste (Water Atomization)	0.41	kg	Landfill (Non-hazardous)
Non-recyclable Powder	0.01	kg	Landfill (Non-hazardous)

Waste Reduction Initiative: The 0.01 kg of non-recyclable powder represents a 35% reduction compared to the previous quarter average of 0.015 kg, attributable to improved sieve mesh optimization.

7.0 Process Parameter Verification

7.1 SLM Equipment Parameters

The EOS M 290 system operated within specified parameters throughout the build cycle. Critical process settings were logged and verified against established protocols.

Parameter	Set Value	Actual Range	Unit	Compliance
Processing Time	13.38	13.38	hours	PASS
Laser Power (Operating)	5.5	5.48-5.52	kW	PASS
System Power (Standby)	3.5	3.48-3.52	kW	PASS
Build Chamber Temperature	80	79-81	°C	PASS
Layer Thickness	0.04	0.039-0.041	mm	PASS

7.2 Gas Management System

Argon consumption was optimized through controlled flow rates and chamber management protocols. Gas usage aligned with calculated requirements based on component geometry and build volume.

Gas Application	Volume	Equivalent Mass	Unit	Purpose
Per Component Processing	54	-	L	Melt pool shielding
Chamber Initial Purge	700	-	L	Oxygen displacement
Total Consumption	-	3.08	kg	Calculated total

8.0 Conclusions and Recommendations

8.1 Quality Compliance Statement

The Selective Laser Melting process for 316L stainless steel flat washers (Batch SLM-WSH-316L-230815-33) has been verified to operate within all specified quality parameters. The 33 produced washers conform to dimensional, material, and finish requirements. Material and energy consumption patterns are consistent with equipment capabilities and process optimization targets.

Final Determination: All inspection criteria satisfied. Batch approved for shipment.

8.2 Process Improvement Opportunities

- Powder Recovery Efficiency:** Current recovery rate of 71.5% presents opportunity for improvement. Recommendation: Evaluate sieve mesh optimization and handling procedures to target 75% recovery.
- Energy Consumption:** While within acceptable ranges, the 64.92 kWh consumption for SLM processing suggests potential for build layout optimization. Suggestion: Review nesting software parameters to reduce laser travel distance.
- Water Usage:** The 16.8 kg water consumption with 97.6% recovery rate demonstrates effective water management. Continue current practices.

8.3 Follow-up Actions

- Schedule preventive maintenance for powder recovery system (Due: September 30, 2023)
- Conduct calibration verification for power monitoring equipment (Due: August 30, 2023)
- Document lessons learned for future similar production runs

Approval Signatures:

Maria Chen
Senior Quality Engineer
Date: August 16, 2023

Robert Takahashi

Production Manager

Date: August 16, 2023

This report constitutes the official quality record for Batch SLM-WSH-316L-230815-33 and shall be retained for a minimum of 7 years per quality management system requirements.